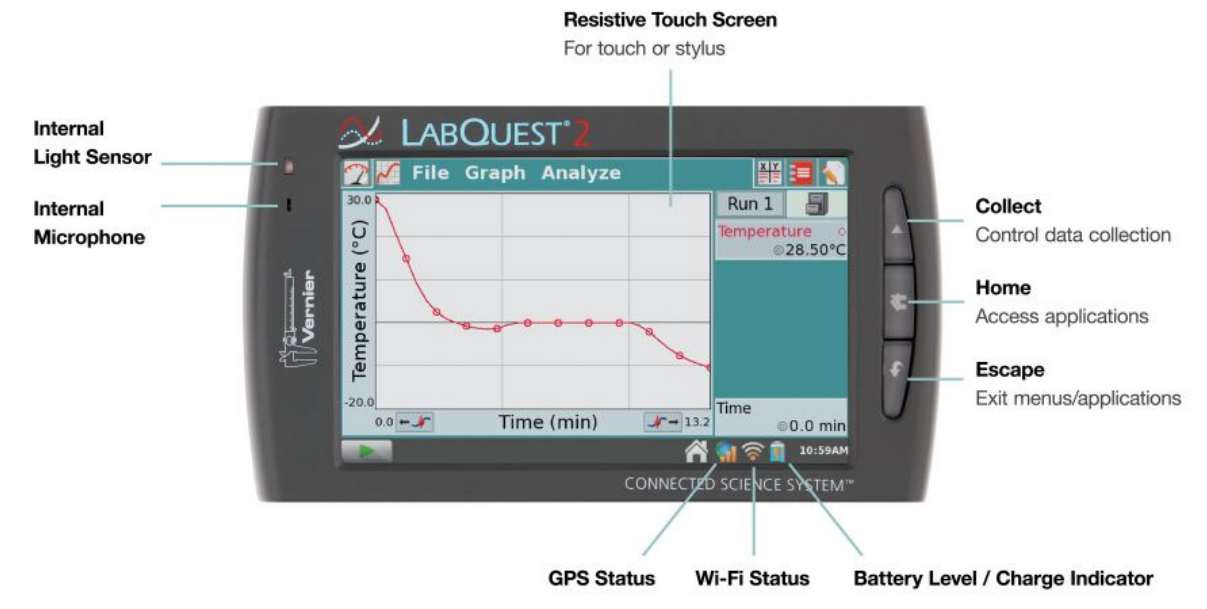
1 MERILNE NAPRAVE

Pri fizikalnih eksperimentih pogosto uporabljamo merilne naprave, s katerimi želimo potrditi teoretične principe in fizikalne pojave. V šolskem prostoru najpogosteje uporabljamo Vernier-jevo merilno napravo LabQuest Hardware “LabQuest® Hardware and Specifications | Vernier” , ki nam omogoča raznovrstne meritve.

1.1 Vernier’s DAQs



Slika 1: Vernier-jeva merilna naprava LabQuest2

Poglejmo si nekaj podatkov o napravi:

- **Display**
  - 11.2 cm x 6.7 cm (13.1 cm diagonal) screen
  - 00 x 480 pixel color display at 188 dpi
  - ED backlight
  - ortrait or landscape screen orientation
  - igh-contrast mode for outdoor visibility
- **Processor**

  - 800 MHz Application Processor

- **Connectivity**
  - Wi-Fi 802.11 b/g/n @ 2.4GHz
  - Bluetooth Smart for WDSS and Go Wireless Sensors

- **User Interface**
  - Resistive touch screen
  - Touch and stylus navigation for efficiency and precision

- **Data Acquisition**
  - 100,000 samples per second
  - 12-bit resolution
  - Built-in GPS, 3-axis accelerometer, ambient temperature, light, and microphone

- **Environmental Durability**
  - Operating Temperature: 0 + 45°C
  - Storage Temperature: -30 + 60°C
  - Splash resistant
  - Rugged enclosure designed to withstand a fall from lab bench

- **Size and Weight**
  - Size: 8.8 cm x 15.4 cm x 2.5 cm
  - Weight: 350 g

- **Ports**
  - 5 sensor channels
  - USB port for sensors, flash drives, and peripherals
  - USB mini port
  - DC power jack
  - MicroSD/MMC slot
  - Audio in and out

- **Storage**
  - 200 MB
  - Expandable with MicroSD and USB flash drive

- **Power**
  - Rechargeable, high-capacity battery

  - DC charging/powering through external adapter (included)

- **cena:**
  - $455

Seveda pa morate dokupiti še senzorje, ki tudi niso cenovno ugodni, saj se njihova cena giblje od $30 … naprej.

1.2 Arduino Data Acquisition System

On the market we can find different DAQ systems which are hi-end products and often expencive (from 100 € .. n k€). Buy we can use Arduino (Uno, nano, …) as low-cost data acqusition system if we do not need scientific accurate data (for pedagogical purposes).

<– Parts of DAQ[[NationalInstruments]] –>

Na plošči vsebuje mikrokrmilnik **Atmega328**, ki lahko opravi podobne naloge, kot smo jih opisali v poglavju **Merilne naprave**.

Poglejmo si nekaj karakteristik tega mikrokrmilnika (Atmel 2017):

- **Advanced RISC Architecture**
  - 131 Powerful Instructions
  - Most Single Clock Cycle Execution
  - 32 x 8 General Purpose Working Registers
  - Fully Static Operation
  - Up to 20 MIPS Throughput at 20MHz
  - On-chip 2-cycle Multiplier
- **High Endurance Non-volatile Memory Segments**
  - 32KBytes of In-System Self-Programmable Flash program Memory
  - 1KBytes EEPROM
  - 2KBytes Internal SRAM
  - Write/Erase Cycles: 10,000 Flash/100,000 EEPROM
  - Data Retention: 20 years at 85°C/100 years at 25°C(1)
  - Optional Boot Code Section with Independent Lock Bits
    * In-System Programming by On-chip Boot Program
    * True Read-While-Write Operation
  - Programming Lock for Software Security
- **Atmel® QTouch® Library Support**

  - Capacitive Touch Buttons, Sliders and Wheels
  - QTouch and QMatrix® Acquisition
  - Up to 64 sense channels

- **Peripheral Features**
  - Two 8-bit Timer/Counters with Separate Prescaler and Compare Mode
  - One 16-bit Timer/Counter with Separate Prescaler, Compare Mode, and Capture Mode
  - Real Time Counter with Separate Oscillator
  - Six PWM Channels
  - ADC
    * 8-channel 10-bit ADC in TQFP and QFN/MLF package
    * Temperature Measurement
    * 6-channel 10-bit ADC in PDIP Package
    * 10-bit Resolution
    * 0.5 LSB Integral Non-Linearity
    * ±2 LSB Absolute Accuracy
    * $13 - 260\mu s$ Conversion Time
    * Up to 76.9kSPS (Up to 15kSPS at Maximum Resolution)
    * Six Multiplexed Single Ended Input Channels
    * Two Additional Multiplexed Single Ended Input Channels (TQFP and VFQFN Package only)
    * Temperature Sensor Input Channel
    * Optional Left Adjustment for ADC Result Readout
    * 0 - VCC ADC Input Voltage Range
    * Selectable 1.1V ADC Reference Voltage
    * Free Running or Single Conversion Mode
    * Interrupt on ADC Conversion Complete
    * Sleep Mode Noise Canceler
  - Two Master/Slave SPI Serial Interface
  - One Programmable Serial USART
  - One Byte-oriented 2-wire Serial Interface (Philips I2C compatible)
  - Programmable Watchdog Timer with Separate On-chip Oscillator
  - One On-chip Analog Comparator
  - Interrupt and Wake-up on Pin Change

- **Special Microcontroller Features**
  - Power-on Reset and Programmable Brown-out Detection
  - Internal Calibrated Oscillator

  - External and Internal Interrupt Sources
  - Six Sleep Modes: Idle, ADC Noise Reduction, Power-save, Power-down, Standby, and Extended Standby

- **I/O and Packages**
  - 23 Programmable I/O Lines
  - 28-pin PDIP, 32-lead TQFP, 28-pad QFN/MLF and 32-pad QFN/MLF

- **Operating Voltage:**
  - 1.8 - 5.5V

- **Temperature Range:**
  - -40°C to 105°C

- **Speed Grade:**
  - 0 - 4MHz @ 1.8 - 5.5V
  - 0 - 10MHz @ 2.7 - 5.5V
  - 0 - 20MHz @ 4.5 - 5.5V

- **Power Consumption at 1MHz, 1.8V, 25°C**
  - Active Mode: 0.2mA
  - Power-down Mode: $0.1\mu A$
  - Power-save Mode: $0.0.75\mu A$ (Including 32kHz RTC)

1.3 Requirements

(Checked is required, unchecked is optional)

Hardware:

☒ Computer
☒ Arduino board (Arduino UNO, Arduino LEONARDO, Arduino NANO, clone products)
☒ basic electronics components for sensors
☐ arduino set starter for example use this kit
☐ arduino sensors set for example use this kit

Software:

☒ Arduino IDE download here
☐ Python (>=3.0)

- ☐ pyserial
- ☐ Ms Excel

Atmel. 2017. “Atmega328p.” 2017. http://ww1.microchip.com/downloads/en/DeviceDoc/Atmel-7810-Automotive-Microcontrollers-ATmega328P_Datasheet.pdf.

“LabQuest® Hardware and Specifications | Vernier.” https://www.vernier.com/products/interfaces/labq/hardware/.